



COLLEGE OF ENGINEERING, CONSTRUCTION, AND LIVING SCIENCES

BACHELOR OF INFORMATION TECHNOLOGY (BIT)

ASSIGNMENT-2, SEMESTER 1, 2026

ANALYSE AND MANAGE PROCESS, STORAGE AND NETWORK MANAGEMENT MODULES
IN OS

ID616001 OPERATING SYSTEMS CONCEPTS

LEVEL 6, CREDITS 15

VERSION 1.0

General Instructions

This assignment contributes 15% to your total course grade.

- This is an individual assessment
- All submissions must be your own original work. If you reference any external resources (online or offline), you must provide appropriate citations and credit.
- You may use online tools or websites (e.g., Google) to assist your work; however, all sources used must be properly acknowledged.
- **Collaboration is strictly prohibited.** All tasks must be completed independently.
- **Assignment Due Date:** [Friday, 12 June 2026, at 11:59 PM](#)

Introduction

This assessment focuses on demonstrating the role of operating systems in the delivery and management of core services that enable network communication, application hosting, and system security. Using a virtualized environment in VMware, you will deploy and manage multiple Linux-based systems to demonstrate how operating system coordinates system resources, security and network services, and how key OS services interact to support reliable network communication and application delivery.

Through configuring routing, deploying web and database services, and implementing firewall rules, you will analyse how operating systems coordinate system resources and service processes.

This assessment directly supports the following learning outcomes:

- **LO1:** Analyse and manage process, memory, storage, and network management — by configuring, monitoring, and securing operating system services across multiple virtual machines.
- **LO2:** Evaluate the design and process models of contemporary operating systems — by examining Linux service management, resource handling, and network process models in a virtualized infrastructure.

System Setup Overview

In this assessment, you will deploy, configure and evaluate an operating system-based service environment using a three-VM network environment within VMware Workstation, consisting of:

- A dual-homed **gateway-vm** acting as the router and firewall.
- Two internal **application servers** (**batcave-vm** and **waynemanor-vm**) running LAMP-based web and database services.

Figure 1 shows the network topology.

Successful completion of this assessment will demonstrate your ability to apply operating system concepts to a real-world scenario, showing an understanding of service management, system resource coordination, and OS-level network functionality.

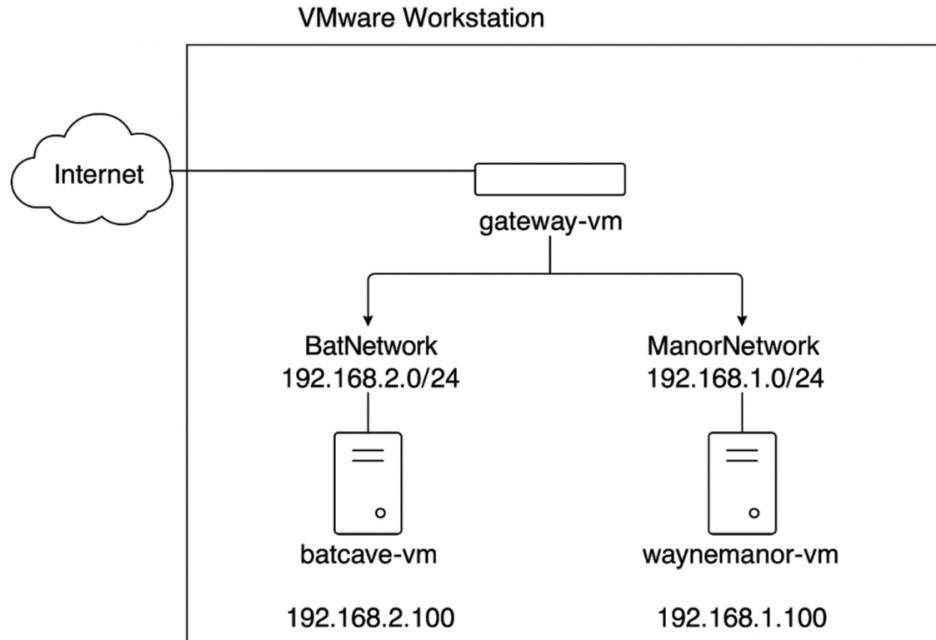


Figure 1: Network Topology

Assessment Context

You are required to deploy and configure three virtual machines within VMware to establish a multi-service Linux environment. The environment will include routing, LAMP-based service deployment, and firewall implementation. The configuration tasks will allow you to demonstrate both conceptual understanding and technical proficiency in managing operating system services.

System Architecture & Network Roles

- **gateway-vm:** Router and firewall
- **batcave-vm:** Internal DB server on BatNetwork
- **waynemanor-vm:** Internal LAMP server on ManorNetwork

Summary of Network Roles

- **gateway-vm:** Dual-homed system with interfaces on BatNetwork, ManorNetwork, and NAT. Provides routing, NAT, and firewall services between subnets and the external network.
- **batcave-vm:** Internal server on BatNetwork hosting the **MySQL database service**. Configured to accept secure connections from the web application server on ManorNetwork through the gateway.
- **waynemanor-vm:** Internal server on ManorNetwork running **Apache and PHP** to serve web applications. Configured to connect to the MySQL service on batcave-vm across subnets.

All virtual machines must be configured with static IPs on their respective internal networks. The NAT interface on the **gateway-vm** must provide Internet access to both internal subnets through routing and masquerading. Note that VMware NAT provides external Internet connectivity, while Linux NAT (iptables masquerading) manages internal routing and subnet access. This setup models a real-world client-server architecture and enables evaluation of inter-process communication, resource management, and OS-level security controls between networked services.

Network Configuration Overview

The assessment requires the creation of a virtualized multi-network environment in VMware Workstation to simulate a small-scale operating system service infrastructure. The setup consists of three Linux virtual machines connected through two internal subnets and one external NAT interface. The **gateway-vm** acts as a router and firewall between the internal networks and the external Internet, facilitating controlled connectivity and inter-service communication between servers.

Virtual Network Configuration

The following VMware virtual networks must be created and assigned to simulate external and internal operating system environments:

- **VMnet8 (NAT):** External interface providing Internet connectivity to the **gateway-vm** through host-based network address translation. This setup allows safe, controlled access to external repositories and prevents direct exposure of the virtual machines to the institutional or public network.
- **VMnet2 (Host-only): BatNetwork** – internal subnet connecting **gateway-vm** and **batcave-vm** (192.168.2.0/24).
- **VMnet3 (Host-only): ManorNetwork** – internal subnet connecting **gateway-vm** and **waynemanor-vm** (192.168.1.0/24).

Each Host-only network must be created using the VMware Virtual Network Editor before assigning network interfaces to the respective virtual machines.

BatNetwork Subnet

The **BatNetwork** subnet connects the **gateway-vm** and the **batcave-vm**, enabling database service communication and management traffic for backend operations.

- Network address: 192.168.2.0
- Subnet mask: 255.255.255.0
- Network prefix: /24
- IP address (batcave-vm): 192.168.2.100
- IP address (gateway-vm on BatNetwork): 192.168.2.10

ManorNetwork Subnet

The **ManorNetwork** subnet connects the **gateway-vm** and the **waynemanor-vm**, supporting application-level communication and web service hosting.

- Network address: 192.168.1.0
- Subnet mask: 255.255.255.0
- Network prefix: /24
- IP address (waynemanor-vm): 192.168.1.100
- IP address (gateway-vm on ManorNetwork): 192.168.1.10

Task 1: VMware and Network Configuration (20 marks)

In this task, you will set up the VMware environment and implement the virtual network topology as defined in the configuration overview. This includes creating and assigning the required virtual networks, configuring network interfaces on each virtual machine, and verifying end-to-end connectivity across all subnets to ensure proper communication between systems.

Requirements:

- Create and assign virtual networks (`VMnet8`, `VMnet2`, `VMnet3`) using the VMware Virtual Network Editor.
- Attach correct network adapters to each VM:
 - `gateway-vm`: NAT, BatNetwork, and ManorNetwork.
 - `batcave-vm`: BatNetwork only.
 - `waynemanor-vm`: ManorNetwork only.
- Configure static IP addresses on all internal interfaces using Netplan.
- Verify connectivity between all VMs through the gateway using ICMP (ping).

Assessment Criteria:

- VMware virtual networks configured correctly (10 marks)
- Network interfaces assigned and IPs configured accurately (5 marks)
- Successful inter-VM connectivity verified (5 marks)

Task 2: Operating System Routing and Service Management (20 marks)

In this task, you will configure system-level routing and Network Address Translation (NAT) to enable communication across the configured subnets. You will also manage and verify key operating system services responsible for routing and network process operations.

Requirements:

- Enable IP forwarding on `gateway-vm`.
- Configure persistent NAT and routing between BatNetwork and ManorNetwork.
- Test routing between `waynemanor-vm` and `batcave-vm`.
- Use OS tools (`ip route show`, `ss`, `ps`, `systemctl`) to analyse routing and process interactions.

Assessment Criteria:

- Functional routing and NAT between internal subnets (10 marks)
- Analysis and explanation of OS-level routing processes (10 marks)

Task 3: Distributed Service Deployment and Management (20 marks)

In this task, you will deploy a distributed web application architecture by configuring separate virtual machines for the database and web servers. You will also manage the relevant operating system services to ensure proper communication and functionality between the components.

Requirements:

- Install and configure **MySQL** on **batcave-vm**.
- Install and configure **Apache** and **PHP** on **waynemanor-vm**.
- Create a simple PHP web page on **waynemanor-vm** that connects to the MySQL database on **batcave-vm**.
- Verify database connectivity and web service access using **curl** or a browser.
- Demonstrate process management (**systemctl**, **ps**, **top**) to show service monitoring and memory/process handling.

Assessment Criteria:

- Web and database services deployed and configured correctly (10 marks)
- Verified inter-service communication across subnets (5 marks)
- Demonstrated understanding of OS process and service management (5 marks)

Task 4: Firewall Configuration and Security Evaluation (20 marks)

In this task, you will implement operating system-level firewall rules using **iptables** and UFW to secure inter-VM communication. You will then evaluate the effectiveness of these configurations in enforcing network access control and system security.

Requirements:

- Configure **IPTables** on **gateway-vm** to provide NAT, port forwarding, and packet filtering.
- Use **UFW** on **batcave-vm** and **waynemanor-vm** to restrict inbound and outbound traffic.
- Ensure only necessary ports are open:
 - HTTP (80) on **waynemanor-vm**
 - MySQL (3306) on **batcave-vm**
- Analyse firewall rule interaction with OS process and network management.
Note: UFW is a front-end interface for iptables that modifies the same underlying kernel-level netfilter rules.

Assessment Criteria:

- **iptables** configuration correct and functional (7.5 marks)
- UFW rules effectively applied on internal servers (7.5 marks)
- Insightful analysis of OS-level security and process interaction (5 marks)

Task 5: System Testing and Validation (10 marks)

In this task, you will conduct functional and connectivity tests to validate routing, distributed service operation, and firewall configurations. You will document the outcomes and confirm that all systems meet the required operational and security objectives.

Requirements:

- Test web application access from `waynemanor-vm` to `batcave-vm` (database query success).
- Verify routing and firewall behaviour by performing traceroute and connection tests.
- Confirm that external Internet access (via NAT) functions as expected.
- Document observed OS-level responses during connection attempts.

Assessment Criteria:

- All services, routing, and firewall rules operate as intended (5 marks)
- Clear testing evidence and troubleshooting documentation (5 marks)

Task 6: Documentation and Reflection (10 marks)

Provide comprehensive documentation linking configurations to OS service and process management principles.

Requirements:

- Include annotated screenshots of key configurations (Netplan, iptables, UFW, routing tables).
- Provide reflection discussing:
 - How NAT and routing enable distributed service delivery.
 - Comparison of iptables and UFW in OS-level control.
 - OS process and memory handling for running services.

Assessment Criteria:

- Documentation clarity, structure, and completeness (5 marks)
- Reflection quality and conceptual depth (5 marks)

Submission Requirements

- Submit a single report (PDF) including configuration details, explanations, and screenshots.
- Append configuration files (e.g., Netplan, IPTables, Apache, MySQL) in the appendix.
- Ensure all figures and outputs are clearly labelled and legible.

Marking Summary

Task	Marks
Task 1: VMware and Network Configuration	20
Task 2: OS Service Management and Routing	20
Task 3: Distributed Service Deployment	20
Task 4: Firewall Configuration and Security Evaluation	20
Task 5: System Testing and Validation	10
Task 6: Documentation and Reflection	10
Total	100 marks